

Beyond Accuracy: Trustworthy Index and Robustness Index for Evaluating CNN

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1. Introduction

We need to explain CNN reliability and XAI from title.

Despite excellent accuracy, a model may not be reliable. Explainability methods, such as Grad-CAM, show that a model may rely mainly on background and irrelevant regions instead of focused objects [? ? ?]. This raises questions about reliability and trustworthiness. **we are not working to improve reliability??**

Traditional CNN model evaluation measures fail to indicate the reliability of the corresponding model.

To address this limitation (**we did not say the limitation yet**), we have proposed two novel evaluation metrics: Trustworthy Index (TI) and Robustness Index (RI) using explainable AI (XAI) methods. We highlight the importance of object reliability (**is this correct? or model reliability??**) as well as accuracy. TI integrates accuracy and object reliability (**??**) in clean images. RI measures the robustness under various perturbations. **need to define robustness with ref** Robustness measures how close the model's accuracy and explanations (object reliance) under perturbation remain to the outputs on clean images (**good but not clear - remember if you have to think to understand a sentence - you should probably rewrite it!**).

The proposed measures TI and RI rely on the fact that *how much a model's predictions rely on real objects instead of the background*. We introduce a new score named LMOS (Localization Mask Overlap Score) to indicate this fact.

As red and yellow regions of the Grad-CAM visualization have stronger influence than other regions for predictions, we consider them as decision-relevant. **why do we need this sentence?**

LMOS defines when the model relies on something to make decisions, what ratio of that reliance falls on focus objects. **not clear**

A model has high accuracy but low LMOS may be affected by shortcut learning, overfitting, bias, and reliance on background features. **can we show this - i.e., a model having high acc but low LMOS and has one or more of the mentioned problems? A model is trustworthy if it has**

ABSTRACT

TBD

high accuracy and high LMOS. Similarly, a model is considered robust under perturbations if it maintains high accuracy stability, and high LMOS stability under perturbations.

need clear distinctions between EI and RI - what EI shows must be different than RI

Models with high TI is considered reliable as both the model accuracy and object reliance are high. Models with moderate to low TI means its accuracy or object focus, or both, are low and limited.

To evaluate the robustness of the models, we have applied nine perturbations (noise, blur, hue, occlusion, etc.) to the test images and then calculated accuracy and LMOS scores after each perturbation.

The mean LMOS stability under a perturbation has been obtained by averaging the image-wise LMOS stability values across all test samples. Then, ERI under a perturbation has been calculated as the mean of both stabilities. Finally, the mean ERI across all perturbations has been calculated and a ranking has been made according to the mean ERI. Like ETI, a model with high ERI indicates that both its accuracy and object focus remain stable under the use of perturbations. A moderate/low ERI reveals that its accuracy, or focus on object, or both change with the use of perturbations.

Two questions that the proposed measures can answer - How trustworthy a model is? Trustworthy means - predictions are based on real objects in the images. How robust a model is? Robustness means - performance of the model will not degrade under perturbations.

To show in exp:

- Acc high but not reliable - may be low robustness or predictions based on BG.
- a model having high acc but low LMOS and has one or more of the mentioned problems
- A model is trustworthy if it has high accuracy and high LMOS - show this. Idea: models with high ACC and high LMOS making more predictions based on objects in images rather than BGs
- Show LMOS still works with perturbations and ACC and other traditional measure fails.

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- Justify distinctions between EI and RI
- Show - traditional measures fail under perturbations but proposed measures hold.

2. Problem Definition

add a table of notations here

3. System Architecture

4. Experimental Setup

4.1. Dataset

We have used four publicly available datasets in our empirical study of the proposed evaluation measures:

- **EPLI**: EPLI (introduced as EgyPLI in [?]) contains 3,588 JPG images of size 256×256 pixels. There are images of 8 species including apple, berry, fig, guava, orange, plum, persimmon, and tomato. The dataset is annotated for image-level classification [?].
- **BDML**: BDML(also known as BDMediLeaves [?]) is a dataset of 2,029 original leaf images (augmented to 38,606) from 10 medicinal plant species. They are captured with an iPhone 13 under natural light at 3024×4032 resolution. This dataset is designed for machine learning-based species identification. We used the original dataset (not the augmented one) in our experiments [?].
- **FLRS**: FLRS (or Flowers [?]) is a large image dataset for flower classification containing 3,670 training images of five flower species including daisies, tulips, sunflowers, dandelions, and roses. It is available through the TensorFlow Datasets (TFDS) collection [?]. what is the resolution of the images in this dataset?
- **KFLR**: This kaggle dataset is very similar to the FLRS dataset above consisting the same five species (daisies, dandelions, roses, sunflowers, and tulips). However, the kaggle version contains 4,242 JPG images of resolution of 320×240 pixels. is this different than the above FLRS?

Specific difference between FLRS and KFLR.

Dataset preparation. The BDML dataset is already split into training, validation and testing sets. We split the other three datasets into training, validation and test subsets with an approximate ratio of 80%, 10% and 10%. Table ?? presents the class-wise training, validation, and test splits. did you preserve the class distributions in the three splits?

did you do this? or already in the sources? To simplify explainability and localization analysis, the test images have been manually annotated to generate corresponding JSON files. These JSON annotations describe the polygonal boundaries of the target leaf in each image. The JSON

Table 1

Class-wise distribution of the experimental datasets.

Dataset / Class	Total	Train	Val	Test
<i>EgyPLI</i>				
Apple	519	415	52	52
Berry	340	272	34	34
Fig	508	406	50	52
Guava	520	416	52	52
Orange	547	437	55	55
Plum	468	374	47	47
Persimmon	527	421	53	53
Tomato	159	127	16	16
Subtotal	3,588	2,868	359	361
<i>Flowers (TFDS)</i>				
Daisy	633	506	64	63
Dandelion	898	718	90	90
Roses	641	512	64	65
Sunflowers	699	560	70	69
Tulips	799	639	80	80
Subtotal	3,670	2,935	368	367
<i>Flowers Recognition</i>				
Daisy	764	611	76	77
Dandelion	1052	842	105	105
Roses	784	627	78	79
Sunflowers	733	586	73	74
Tulips	984	787	98	99
Subtotal	4,317	3,453	430	434

files have been converted into binary masks. Here, each mask represents only the target leaf region and excludes the background and other objects in the scene.

4.2. Baseline models and evaluation metrics.

TBD

4.3. Classification Performance

4.4. Implementation of the Proposed Metrics

Both of the proposed TI and RI metrics use LMOS score. LMOS measure the consistency between model-generated explanations and ground-truth object regions. One liner on TI and RI to be added here We briefly describe the implementation details of these metrics below.

4.4.1. Localization Mask Overlap Score (LMOS).

need to say LMOS is an instance specific (or local) measure. Other metrics e.g., acc - is a global measure - indicate overall perf. Thats why we use $\overline{LMOS}$ in TI definition, just need to say it outloud here LMOS calculates how much a model focuses on true object regions (instead of backgrounds) for prediction. Let $H \in \mathbb{R}^{h \times w}$ denote the Grad-CAM heatmap generated from the last convolutional layer of a network. H is resized to the model's input image size. if H is size how can it be $\mathbb{R}^{h \times w}$? The resized heatmap size is 299×299 for Xception, 224×224 for other models.

A binary attention map is obtained from H by thresholding:

$$\hat{H}(i, j) = \mathbb{I}(\text{Resize}(H)(i, j) > \tau)$$

where $\mathbb{I}(\cdot)$ is an indicator function and τ is the threshold to or for??. We set $\tau = 0.7$ why? through experiments or what? how $\tau = 0.7$ restricts to yellow and red pixels?? to restrict our LMOS calculations to red and yellow pixels, as the red and yellow pixels exhibit the highest activation levels in model predictions. Hence, $\hat{H}(i, j)$ is a binary mask representing the activated decision-relevant regions.

LMOS is defined as the ratio of activated decision-relevant regions inside the object region and the total activated decision-relevant regions. Mathematically:

$$\text{LMOS} = \frac{\sum_{i=1}^h \sum_{j=1}^w \hat{h}(i, j) M(i, j)}{\sum_{i=1}^h \sum_{j=1}^w \hat{H}(i, j)} \quad (1)$$

where $M \in \{0, 1\}^{h \times w}$ denote the resized ground-truth binary segmentation mask. waht is L here? what is $\hat{h}$ here? how do we get M???

4.4.2. Trustworthy Index (TI)

In this article, we propose Trustworthy Index (TI) to measure trustworthiness of a CNN model's predictions. We combine the accuracy, explainable consistency and the mean LMOS (Eq. ??) to define TI.

Let acc denote the classification accuracy of a model on test set and $\overline{\text{LMOS}}$ denote the mean LMOS scores across the same test samples. We define TI defined as follows:

$$\text{TI} = \frac{acc + \overline{\text{LMOS}}}{2}.$$

In this study, both acc and LMOS are equally important and are normalized to the range $[0, 1]$. acc measures predictive performance, whereas LMOS measures the mean explanation consistency with the target object. We provide equal weighting to both components to avoid favoring either predictive performance or explanatory quality. need to rewrite

Variations of TI. To demonstrate the impact of LMOS and acc on the proposed TI, we have calculated two variants of TI as folllows:

$$\text{TI}_{\text{w/o LMOS}} = \frac{acc}{2}, \quad \text{TI}_{\text{w/o acc}} = \frac{\text{LMOS}}{2}.$$

4.4.3. Robustness Index (RI)

We estimate the robustness of a model by measure its' performance deviation under purturbation in the test datasets. Let acc_c denote the accuracy of a model on a clean test dataset (not purturbed) and acc_p denote the accuracy obtained on the test set after applying a specific perturbation p (see Sec. ?? for all the purturbation methods used in this research). We intorduce the Accuracy Stability (S_{acc}) score if not introduced by you cite under a specific perturbation p as:

$$S_{acc}^p = 1 - \frac{|acc_c - acc_p|}{acc_c + acc_p + \epsilon}$$

where ϵ is a small positive constant introduced to avoid division by zero error. A higher value of S_{acc}^p indicates that the model's performance is more consistent under the corresponding perturbed condition p .

To estimate an overall measure of S_{acc} , we compute S_{acc}^p across all $p \in P$ where P is the set of all purturbation methods used in the experiments, see Section ?? and take the mean of all S_{acc}^p . This can be defined as:

$$\overline{S}_{acc} = \frac{1}{|P|} \sum_{p=1}^{|P|} S_{acc}^p$$

Similarly, we introduce LMOS Stability to measures how closely a model's object reliance under perturbation remains to its object reliance on clean images. To evaluate the robustness under input perturbations, we have measured the stability of LMOS when input images are perturbed. need rewriting

For each test sample, we have computed LMOS for both the original image and its perturbed version. Let N be the number of test samples in a test set. Let $\text{LMOS}_{\text{clean}}$ denote the LMOS value computed from an original image and $\text{LMOS}_{\text{perturbed}}$ denote the LMOS value computed from the corresponding perturbed image.

To quantify the consistency between these two values, we define the LMOS stability under a perturbation:

$$S_{\text{LMOS}} = 1 - \frac{|\text{LMOS}_{\text{clean}} - \text{LMOS}_{\text{perturbed}}|}{\text{LMOS}_{\text{clean}} + \text{LMOS}_{\text{perturbed}} + \epsilon}$$

, where ϵ is a small constant to avoid division by zero error. why not the denominator is using abs values?

A higher value of S_{LMOS} indicates that the model's reliance on the focus object remains more consistent under the corresponding perturbation. The LMOS stability under a perturbation p is calculated as an average across all N test samples as:

$$S_{\text{LMOS}}^{(p)} = \frac{1}{N} \sum_{i=1}^N S_{\text{LMOS},i}^{(p)} \quad (2)$$

where $S_{\text{LMOS},i}^{(p)}$ denotes the LMOS stability of the i -th test sample under the perturbation p , and N represents the total number of test samples.

To obtain an overall measure of LMOS stability across all evaluated perturbations, the perturbation-specific LMOS stability scores are subsequently averaged as:

$$\overline{S}_{\text{LMOS}} = \frac{1}{P} \sum_{p=1}^P S_{\text{LMOS}}^{(p)}. \quad (3)$$

where P denotes the total number of evaluated perturbations. In this study, $P = 9$.

To jointly evaluate the robustness of both classification performance and explainability under input perturbations,

we have proposed the *Explainable Robustness Index* (ERI). Unlike accuracy stability and LMOS stability, which separately assess the consistency of predictive performance and model localization behavior, ERI provides a unified measure of robustness by combining the accuracy stability and LMOS stability.

Let s_{acc}^p be accuracy stability and s_{LMOS}^p be LMOS stability under perturbation p . Now, ERI under perturbation p is calculated as:

$$ERI^p = \frac{s_{acc}^p + s_{LMOS}^p}{2}. \quad (4)$$

Let $\bar{S}_{acc}$ denote the overall accuracy stability and $\bar{S}_{LMOS}$ denote the overall LMOS stability, both obtained by averaging their corresponding perturbation-specific stability scores across the P evaluated perturbations. The proposed ERI is defined as:

$$ERI = \frac{\bar{S}_{acc} + \bar{S}_{LMOS}}{2}. \quad (5)$$

Here, $\bar{S}_{acc}$ represents the overall consistency of classification accuracy between clean and perturbed inputs, while $\bar{S}_{LMOS}$ represents the overall consistency of the model's reliance on the focus object under perturbations. Therefore, ERI jointly captures two complementary dimensions of robustness: *predictive stability* and *explanatory stability*.

Since both component scores are bounded in the same range, the resulting ERI also provides a normalized measure of overall robustness. A higher ERI indicates that a model simultaneously maintains its classification performance and preserves its object-focused behavior across the evaluated perturbations. Conversely, a lower ERI indicates that the model exhibits greater instability in one or both dimensions. **Variations of RI.** To analyze the contribution of the individual components of RI (i.e., S_{acc} and S_{LMOS}), we calculate two variants by removing one component at a time:

$$RI_{w/o \text{ LMOS}} = \frac{\bar{S}_{acc}}{2}, \quad RI_{w/o \text{ acc}} = \frac{\bar{S}_{LMOS}}{2}.$$

4.5. Grad-CAM Visualization

Grad-CAM visualization shows that CNN models often predict correctly but they sometimes rely on the background or non-object regions. Besides, under perturbations, their reliance on object pixels hugely changes. [it is not necessary here. We may include the implementation of grad-cam in the method implementation section above](#)

4.6. Perturbation methods

To evaluate the robustness of the proposed framework under distribution shifts and common image corruptions, we have applied nine image perturbation methods to the test set. [if anything in the table is missing below, add it in the corresponding bullet](#) The perturbation parameters have been fixed across all experiments to ensure reproducibility and

enable fair comparisons among different CNN architectures. The details of the perturbations are given below: [missing references here. Add 1 or more references to papers applying these for the same reason? i.e., evaluation under perturbation](#)

- **Gaussian Blur:** Applied using a 5×5 kernel to simulate the loss of fine spatial details caused by defocus or low-quality imaging conditions.
- **Gaussian Noise:** Additive Gaussian noise with zero mean and standard deviation 0.05 has been introduced to model sensor noise and random intensity fluctuations.
- **Brightness Adjustment:** Increased by adding a constant value of 0.2 to normalized pixel intensities, after which values have been clipped to the valid range $[0, 1]$.
- **Hue Shift:** Hue values in the HSV color space have been shifted by 15 units while preserving the saturation and value channels, simulating color variations caused by illumination or camera characteristics.
- **Contrast Adjustment:** Modified using mean-centered linear scaling: $I' = (I - \mu_I) \times 0.5 + \mu_I$, where I denotes the input image and μ_I represents the mean pixel intensity (computed per channel). The resulting values have been clipped to the range $[0, 1]$.
- **Gamma Correction:** Nonlinear illumination changes have been simulated using gamma correction: $I' = I^{2.0}$, where pixel intensities have been raised to the power of $\gamma = 2.0$.
- **Motion Blur:** Simulated using a 7×7 horizontal averaging kernel to emulate camera or object motion during image acquisition.
- **JPEG Compression:** Artifacts have been introduced by encoding and decoding the images with a quality factor of 30, simulating degradation due to lossy storage or transmission.
- **Random Occlusion:** A randomly positioned rectangular region corresponding to 20% of the image height and 20% of the image width has been masked by setting its pixel values to zero, thereby simulating partial object occlusion.

4.7. Training Configurations

We used TensorFlow/Keras on the Kaggle [cite kaggle](#) environment with two NVIDIA Tesla T4 GPUs to training and evaluation of all the CNN models in our experiments. The training configuration has followed the settings adopted in the original EgyPLI study[?]. [what are these? please specify. is it only for EgyPLI or all 4 datasets??](#) The custom CNN was trained from scratch, while other models have been initialized with ImageNet-pretrained weights [cite here](#) and their backbone layers have been frozen.

For the pretrained models, the original classification heads have been removed and replaced with a common classification head consisting of Global Average Pooling [cite](#), a dropout layer [cite](#) with dropout rate 0.3, a fully connected layer with 128 ReLU units [cite](#), and the final softmax layer [?] has been configured according to the number of classes in each dataset. The Adam optimiser [cite](#) has been used with a learning rate of 5×10^{-5} and categorical cross-entropy loss [arg1](#) in all model optimizations.

5. Result and Discussion

5.1. Grad-CAM Visualization

5.1.1. Threshold Justification

5.2. Explainable Trustworthiness Index (ETI)

5.3. Explainable Robustness Index (ERI)

EgyPLI. XXXXXX

5.3.1. *BDMediLeaves*

5.3.2. *Flowers*

5.3.3. *Flowers Recognition KAGGLE Dataset*

5.4. Cross-Dataset Analysis

6. Related Works

7. Conclusion

Data and Code availability

Supplementary code and data (including .json files and .keras files) associated with this research are available at:

https://github.com/ragib1996/ETI_ERI

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Disclosure statement

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References